



Project Brief: Artisan Service Marketplace (MVP)

Project Lead: Adepoju Oluwatobi **Timeline:** 8 Weeks (Accelerated) **Core Objective:** To build a hyper-local marketplace where artisans negotiate custom job prices and individuals pay via an escrow-backed wallet system with an automated 5% platform commission.



Technical Stack

- **Frontend:** Next.js 15 (App Router)
- **Backend:** Node.js (Express)
- **Database:** MongoDB (utilizing ACID Transactions for financial integrity)
- **Authentication:** KroxT Secure System
- **Payments:** Paystack
- **Notifications:** Gmail and in app notification



Team Structure

- **Lead Engineer:** System Architecture, KroxT Auth deployment, and lead on final Production Hosting.
- **2 Frontend Developers:** Responsible for the Customer Interface (Search, Wallet UI) and the Artisan Interface (Portfolios, Negotiation/Quote tool).
- **3 Backend/DevOps Developers:** Responsible for the Financial Engine (Escrow, 5% Split), Marketplace Logic (Bidding, Chat), and the infrastructure (CI/CD, MongoDB optimization, and Cloud scaling).



8-Week Accelerated Roadmap

Phase 1: Identity & Foundation (Weeks 1–2)

- **KroxT Deployment:** Establish secure, multi-role authentication for Artisans and Customers.
- **Fast Onboarding:** Implement a frictionless signup flow (Email/Phone) to bypass heavy NIN/BVN requirements for the MVP.
- **DB Architecture:** Design MongoDB collections for Users, Jobs, Wallets, and Negotiated Quotes.

Phase 2: Negotiation & Search (Weeks 3–4)

- **Discovery Engine:** Build the search and filtering logic to find artisans by skill and proximity.
- **The Quote System:** Create the API for Artisans to enter custom prices based on materials, labor, and distance.
- **Wallet Funding:** Integrate payment gateways to allow users to fund their in-app wallets.

Phase 3: Transactional Escrow (Weeks 5–6)

- **The "Lock" Mechanism:** Upon job acceptance, the system "freezes" the agreed amount in the customer's wallet.
- **The 95/5 Split:** Upon job completion sign-off, trigger the automated payout:
 - **95%** to the Artisan's balance.
 - **5%** to the Platform Revenue account.
- **Financial Integrity:** Ensure all wallet movements are wrapped in MongoDB transactions to prevent data loss.

Phase 4: Trust & Production (Weeks 7–8)

- **Feedback Loop:** Mutual 5-star rating system for both parties to ensure platform quality.
- **Alerts & Messaging:** Finalize Termii integration for SMS job alerts and real-time in-app chat.
- **Production Deployment:** Transition to live API keys and secure cloud hosting.

Key Project Requirements

1. **Negotiated Pricing:** No fixed rates; the price is agreed upon by both parties before the job begins.
2. **Escrow Protection:** Funds are proven and locked before the artisan starts work to prevent payment defaults.
3. **Revenue Model:** An automated 5% commission is deducted from every completed transaction.